

Imad Saddik

📍 Meknès, Morocco | 🌐 imadsaddik.com | ✉️ simad3647@gmail.com | ☎️ +212 614271986 | 🐙 @ImadSaddik | 🌐 @ImadSaddik

Work experience

AGC Glass Europe | Full Stack Engineer

Jun. 2024 - Present | Remote

- **Enterprise AI Chatbot:** Designed and scaled an internal agentic RAG chatbot using Elasticsearch to serve thousands of daily users; optimized backend systems for high availability, added keyboard-navigable web accessibility, and engineered an agentic loop connecting the iTop ticketing API for autonomous IT support.
- **CI/CD & Security Optimization:** Spearheaded backend and deployment infrastructure upgrades within GitLab CI/CD, drastically reducing pipeline build times from 40+ minutes while integrating Dynamic Application Security Testing (DAST) to scan dependencies.
- **FRG Multi-lingual Extraction:** Built a production-ready data migration pipeline using LLMs and Vision-Language Models (VLMs) to transcribe and extract 30+ complex metadata columns from 1,500+ multi-page scanned, multi-lingual PDF reports with a 95% validation accuracy rate.
- **Custom Search Architecture:** Architected a custom explainable search and filtering interface using an LLM to classify query relevance (High/Medium) with written reasoning, explicitly bypassing vector database scoring limitations to meet strict client precision constraints.
- **Upstream Order Automation:** Developed an automated purchase order ingestion pipeline running every 15 minutes that leverages LLMs and VLMs to extract multi-format items, cross-references them against an internal Excel mapping table, and directly transmits structured data to SAP.
- **Email Product Stock Assistant:** Built an automated customer inquiry system via SMTP and LLMs to parse product requests, automatically query internal stock systems, and compile structured availability reports for the sales department.

AI Inside SARL | Software & AI Engineer

Jun. 2024 - Present | Kenitra, Morocco

- **Multi-Depot Vehicle Routing:** Developed a time-windowed vehicle routing optimization system using OSRM to pre-compute travel subgraphs stored in PostgreSQL; implemented a circular-expansion clustering algorithm to automate bus fleet optimization and built a 2D map visualization interface in Vue.js.
- **Digital TPM Pipeline:** Digitized a paper-based lean manufacturing red tag workflow by building a Vue.js/Python tablet application utilizing SQLite with Write-Ahead Logging (WAL) for concurrent safety; designed a custom path-mapping layout for technicians and a gamified plant floor TV dashboard.
- **Industrial Piece Detection:** Trained and deployed deep learning computer vision models within a physical inspection chamber to identify unlabeled manufacturing parts based on high-confidence probability scores, directly cutting inventory losses.

Open source & technical community

freeCodeCamp & 3CodeCamp | Technical Educator & Content Creator

2024 - Present

- **LLM Architecture From Scratch:** Authored and produced a comprehensive course demonstrating how to build and train a 50M-parameter Transformer model from scratch using PyTorch, covering tokenization, tensor debugging, mathematical early stopping, and Supervised Fine-Tuning (SFT).
- **Embedding Models Benchmarking:** Developed a widely-shared course on evaluating embedding models using custom evaluation metrics (Recall@K, NDCG@K, MRR) against a 1,000-chunk semantic-chunked golden dataset.
- **Production Deployment Course:** Created a 20+ hour curriculum covering advanced production systems engineering, detailing Unicorn/Supervisor process management, Nginx reverse proxy tuning, Cloudflare CDN integration, Nmap vulnerability scanning, and automated GitHub Actions CI/CD pipelines (SAST/DAST).

Independent Projects

- **Setuprkhis.com (Price Tracker):** Built a highly scalable Moroccan e-commerce search engine monitoring 100,000+ products across 20+ stores; migrated infrastructure from SQLite to PostgreSQL to eliminate concurrent write locking, utilized asynchronous HTTP parallelization to drop scraping times under 3 hours, and ran local batch LLM inference for 40 hours to standardize multi-lingual product features.
- **DoCamp Mobile App:** Engineered an early-stage native Android RAG application in Java using a custom-built local SQLite vector database and a from-scratch k-Nearest Neighbors (k-NN) similarity search algorithm to handle document chunk context processing.
- **SFT Bodmagh Dataset:** Manually curated, structured, and open-sourced a high-quality Supervised Fine-Tuning dataset on Hugging Face for the Moroccan Darija language to resolve scarcity in regional dialect training sets.

Technical skills

- **Languages:** Python (Async, PyTorch), Java, JavaScript, TypeScript, Bash, SQL.
- **Frameworks & Tools:** FastAPI, Django, Vue.js, React.js, Unicorn, Supervisor, Nginx.
- **Data Systems:** Elasticsearch, PostgreSQL, Meilisearch, SQLite (WAL mode), Redis.
- **AI & Inference:** llama.cpp, llama-swap, vLLM, VLMs, Agentic RAG, LLMs, Fine-tuning.
- **Infrastructure & Security:** Ubuntu Server, Docker, Git, GitHub Actions, GitLab CI/CD, DAST/SAST, Cloudflare.

Education

- **State Engineering Degree (Equivalent to Master's) in Industrial Engineering**
ENSAM Meknès | Specialization: AI & Data Science | Mention Très Bien (Highest Honors)
- **University Diploma of Technology (DUT) in Electrical Engineering**
ESTM Meknès | Mention Très Bien (Highest Honors)